Impact assessment of typhoon-induced population confinement on multi-pathogen of respiratory diseases dynamics in Hong Kong

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Background

Climate change is increasing extreme weather events, yet their impacts on multi-pathogen transmission remain poorly understood. We used the September 2025 typhoon lockdown in Hong Kong as a natural experiment to quantify its effects on six co-circulating pathogens using a Bayesian multi-strain SEIRS model. The same population-wide disruption produced heterogeneous outcomes: SARS-CoV-2 showed the largest transmissibility reduction (37.4%), while influenza A and RSV showed moderate declines and influenza B was unaffected. Epidemic timing and cross-immunity further shaped pathogen competition. Our findings reveal that extreme weather events can differentially reshape multi-pathogen dynamics and inform epidemic preparedness under climate change.

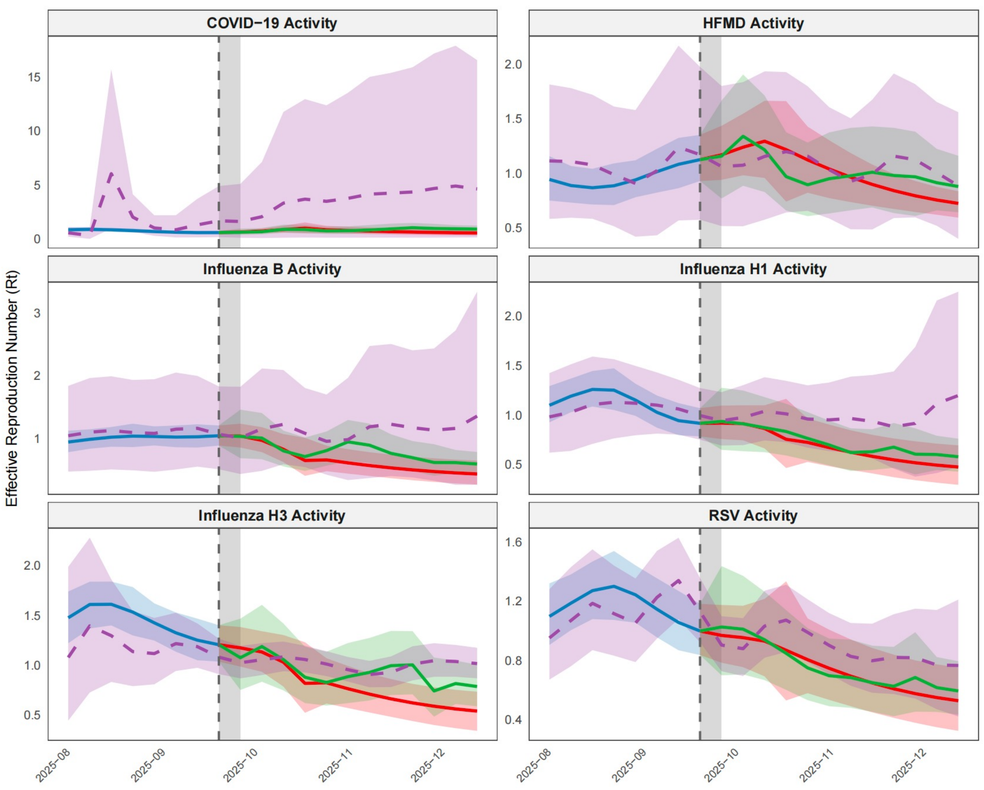
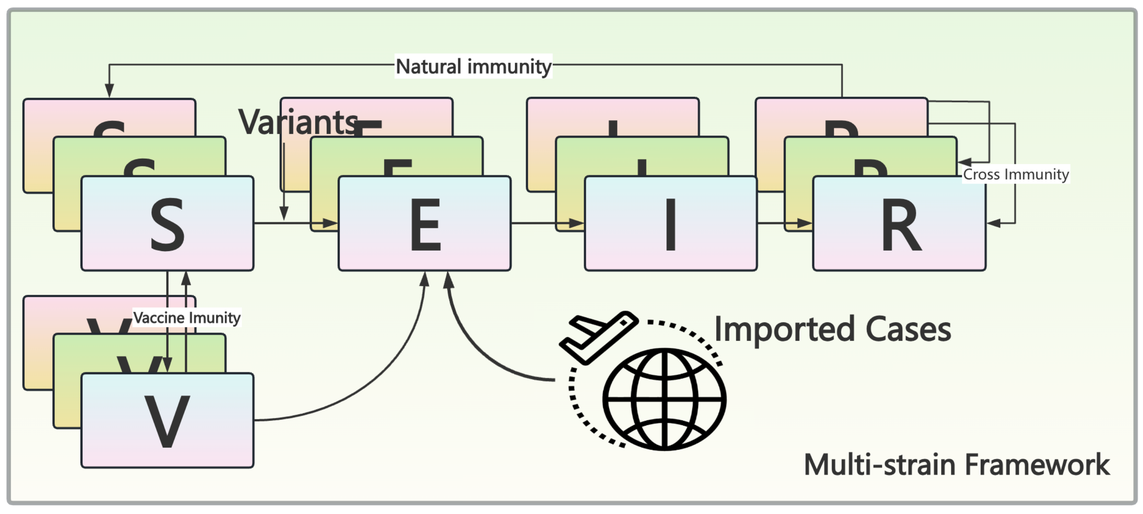


Figure 2. Comparison of Rt Estimates Across Different Models. (Green: with typhoon effects; Red: without typhoon effects; Purple dashed: EpiEstim; Grey shading: typhoon period)

Methods

We developed a data-driven multi-strain forecasting model integrating natural and vaccine-induced immunity, public health interventions, 6 variants, and international importation risks across 15 countries.



Mobility: $\Delta_i = \frac{1}{N_i}\sum_{k\neq i}(M_{ik}I_k - M_{ki}I_i)$

Genetic diversity: $\beta_t = \beta_{spline} \cdot \exp\left(\sum_{v=1}^{V} \delta_v \cdot p_{t,v}\right)$

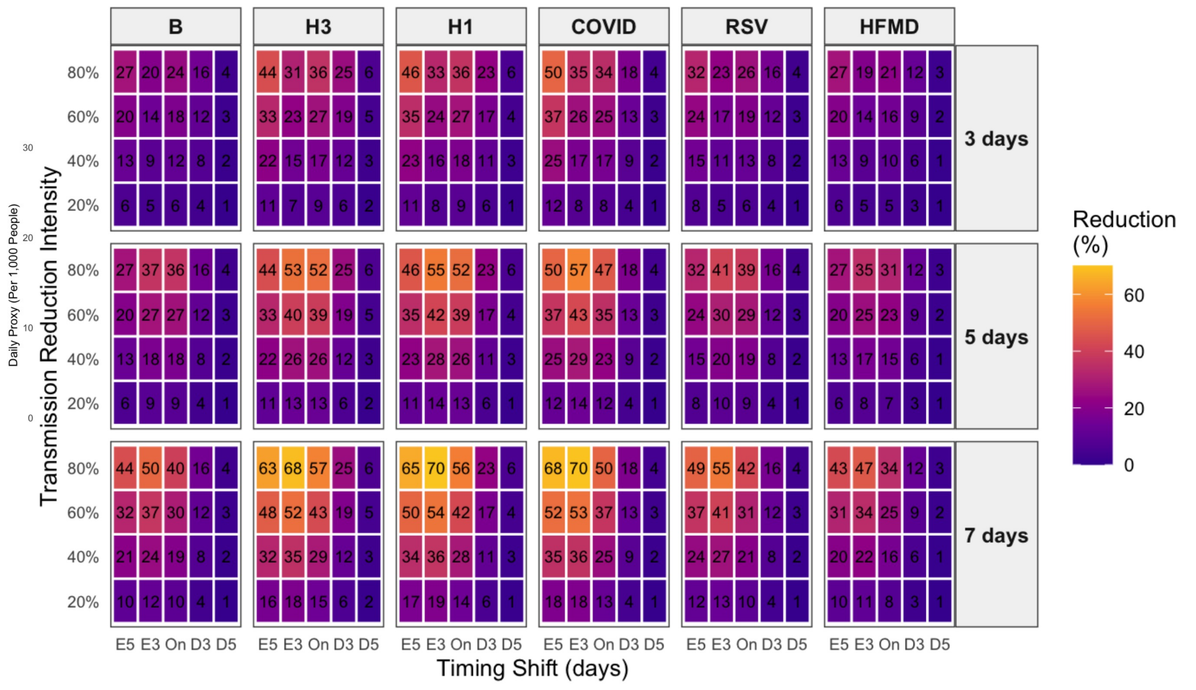


Figure 3. Impact of Typhoon Timing and Duration on Epidemic Dynamics

Results

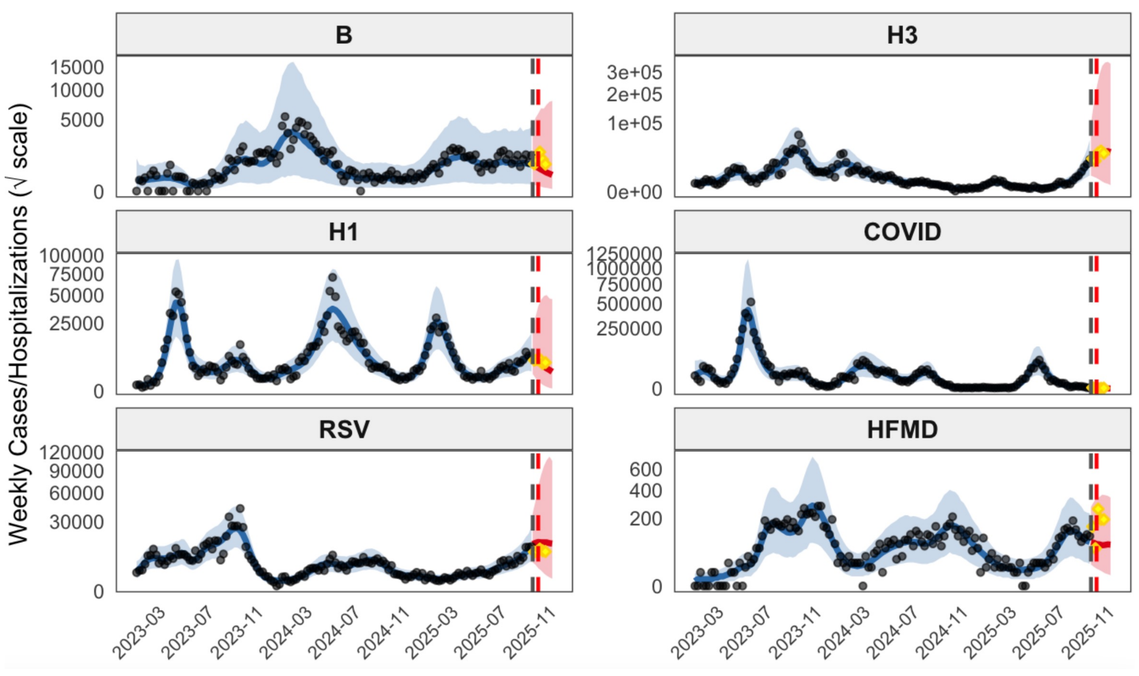


Figure 1. Fitting and Forecasting Results for Multi-strain Dynamics.

Discussion

Extreme weather events reshape multi-pathogen ecosystems rather than simply suppress transmission. Our study reveals pathogen-specific and timing-dependent effects of typhoon-induced disruption, providing a new framework for climate-informed epidemic forecasting and intervention optimization.

References

[1] Kramer SC, et al. Characterizing the interactions between influenza and respiratory syncytial viruses and their implications for epidemic control. *Nat Commun*. 2024;15:10066.

Acknowledgements

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